

Engine



The 3.7L (4V) is a (TiVct) V6 engine with the following features:

- Dual overhead camshafts
- Four valves per cylinder
- Sequential Multi-Port Fuel Injection (SFI)
- A composite lower intake manifold and upper intake manifold
- Aluminum cylinder heads
- An aluminum, 60-degree V-cylinder block
- Twin independent Variable Camshaft Timing (VCT) system
- The electronic ignition system with 6 ignition coils
- 305 horsepower

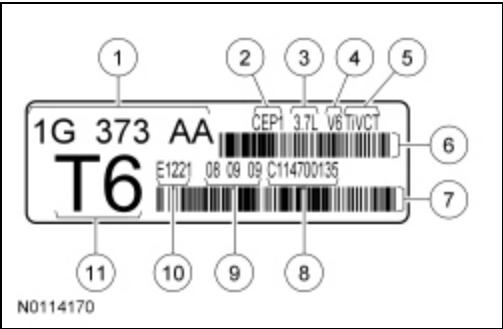
Engine Identification

For quick identification, refer to the safety certification decal.

- The decal is located on the LH front door lock face panel.

Engine Code Information Label

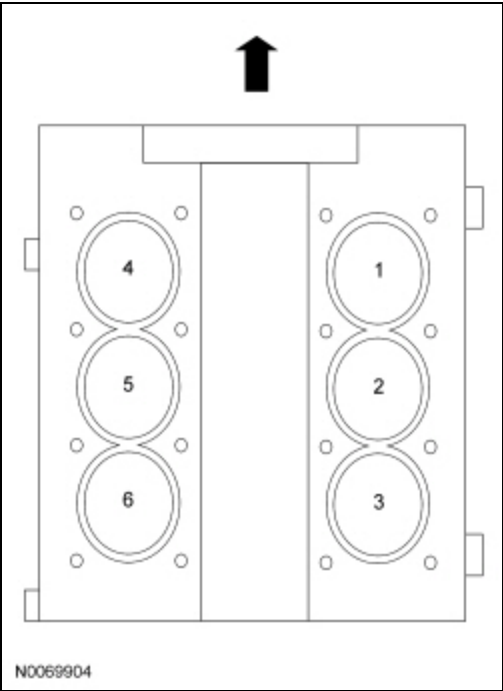
The engine code information label, located on the front side of the valve cover, contains the following:



Item	Description
1	Engine part number
2	Engine plant (Cleveland)
3	Engine displacement
4	Engine configuration
5	Twin independent VCT system
6	Bar code
7	Bar code
8	Running number
9	Engine build date (DDMMYY)
10	Plant shift line

Item	Description
11	Derivative code

Engine Cylinder Identification



Exhaust Emission Control System

Operation and required maintenance of the exhaust emission control devices used on this engine is covered in the Powertrain Control/Emissions Diagnosis (PC/ED) manual.

Induction System

The SEI provides the fuel/air mixture needed for combustion in the cylinders. The 6 solenoid-operated fuel injectors:

- are mounted between the fuel rail and the intake manifold.
- meter fuel into the air intake stream in accordance with engine demand.
- are positioned so that their tips direct fuel just ahead of the engine intake valves.

Valve Train

The valve train uses Direct Acting Mechanical Buckets (DAMB) . The camshaft lobes are positioned directly above mechanical buckets which are positioned on top of the valves.

Twin IndependentVariable Camshaft Timing (VCT) System

The twin independent VCT system allows variable control of the valves that optimizes combustion at full load providing improved power and low speed torque (broadening the torque curve) which enables variable valve overlap which provides better fuel economy and emissions and provides optimized cold start operation with improved exhaust emissions.

PCV System

All engines are equipped with a closed-type PCV system recycling the crankcase vapors to the upper intake manifold.

Lubrication System

The engine lubrication system is of the force-feed type in which oil is supplied under full pressure to the crankshaft, connecting rod bearings, timing chain tensioners and VCT solenoids. The flow of oil to the valve tappets and valve train is

controlled by a restricting orifice located in the cylinder head, valve train oil feed tubes. Die cast aluminum deep sump oil pan for 10,000 mile oil change intervals.

Oil Pump

The lubrication system is designed to provide optimum oil flow to critical components of the engine through its entire operating range.

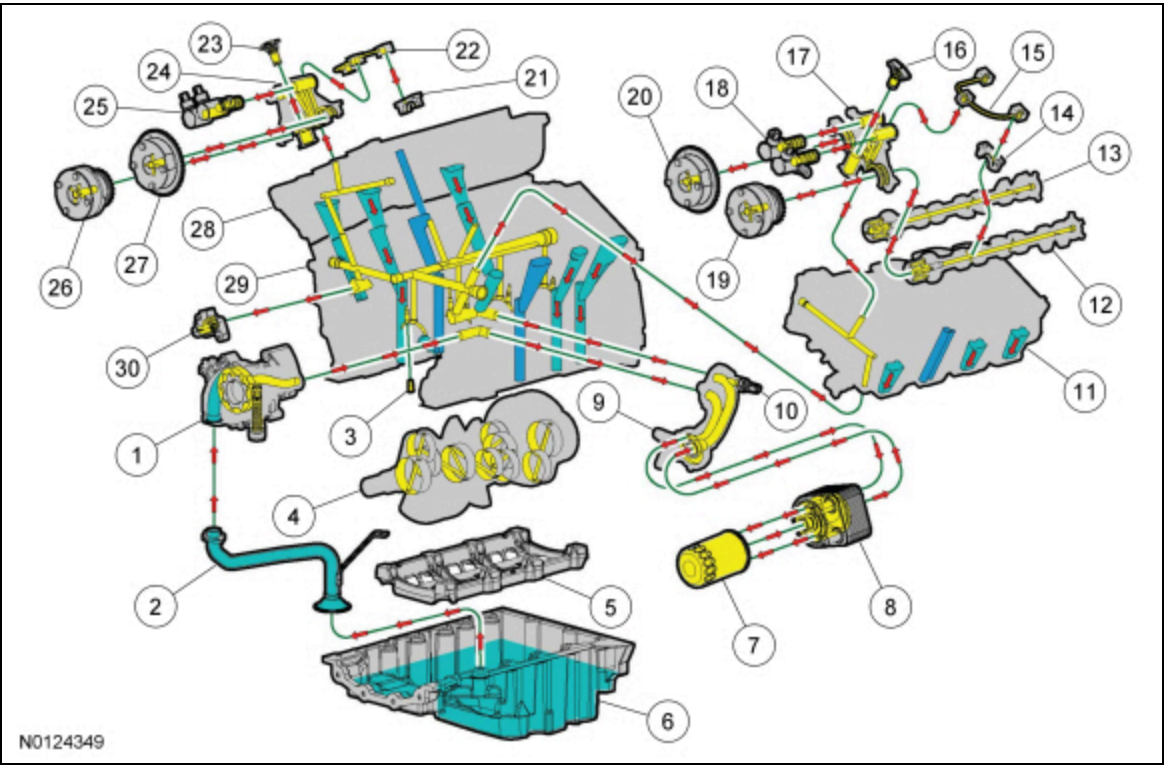
The heart of the system is a positive displacement internal gear oil pump.

Generically, this design is known as a gerotor pump, which operates as follows:

- The oil pump is mounted on the front face of the cylinder block.
- The inner rotor is piloted on the crankshaft post and is driven through flats on the crankshaft.
- System pressure is limited by an integral, internally-vented relief valve which directs the bypassed oil back to the inlet side of the oil pump.
- Oil pump displacement has been selected to provide adequate volume to ensure correct oil pressure both at hot idle and maximum speed.
- The relief valve calibration protects the system from excessive pressure during high-viscosity conditions.
- The relief valve is designed to provide adequate connecting rod bearing lubrication under high-temperature and high-speed conditions.

Engine Oil Flow Illustration

Engine Assembly



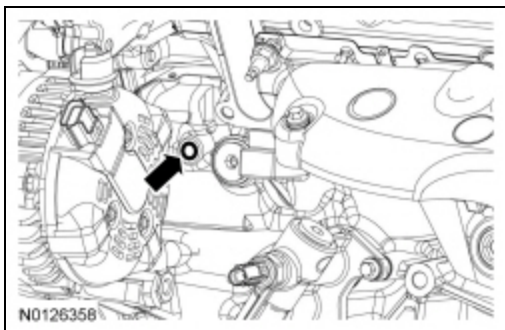
Item	Part Number	Description
1	66006600	Oil pump
2	66226622	Oil pump screen and pickup tube
3	—	Piston oil cool valve (part of 6010)
4	63036303	Crankshaft
5	6C3646C364	Engine main bearing cap support
6	66756675	Oil pan
7	67316731	Oil filter

8	6A642 6A642	Oil cooler
9	6881 6881	Oil filter adapter
10	9278 9278	Engine Oil Pressure (EOP) switch
11	6049 6049	LH cylinder head
12	6250 6250	Exhaust camshaft
13	6250 6250	Intake camshaft
14	6A258 6A258	LH camshaft cap
15	6K602 6K602	LH valve train oil feed tube
16	6K254 6K254	LH secondary timing chain tensioner
17	6B280 6B280	LH camshaft mega cap
18	6M280 6M280	LH intake and exhaust camshaft Variable Camshaft Timing (VCT) oil control solenoid
19	6C525 6C525	LH exhaust camshaft VCT
20	6256 6256	LH intake camshaft VCT
21	6A280 6A280	RH camshaft cap
22	6K602 6K602	RH valve train oil feed tube
23	6C270 6C270	RH secondary timing chain tensioner
24	6B280 6B280	RH camshaft mega cap
25	6M280 6M280	RH intake and exhaust camshaft VCT oil control solenoid
26	6C525 6C525	RH exhaust camshaft VCT
27	6256 6256	RH intake camshaft VCT
28	6049 6049	RH cylinder head
29	6010 6010	Cylinder block
30	6K254 6K254	Primary timing chain tensioner

Cooling System

The engine cooling system includes the following:

- Radiator
- Coolant pump
- Electric fan assembly(s)
- Degas bottle (aids in maintaining the correct volume of engine coolant)
- Coolant thermostat
- Coolant hoses
- Engine oil cooler



The channel cover plate is located under the engine front cover mounted to the block. It contains a timing chain idler and press and place gaskets separating engine coolant and engine oil. A weep hole is provided on the front left side of the engine behind the generator. If oil or coolant is leaking from the weep hole, then a gasket has failed and must be replaced.